

Scaling Ensembles on Harder Data: Function-Space Similarity Across Parameter Count

Experimental notes generated from the `scaling-ensembles` repository

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Abstract

This report studies whether independently trained neural networks become more functionally similar as their parameter count increases. The starting point is Fort, Hu, and Lakshminarayanan’s “Deep Ensembles: A Loss Landscape Perspective”, which argues that deep ensembles work well because independent random initializations explore distinct modes in function space, while trajectory-local subspace methods tend to remain inside one prediction-space mode. Here we reproduce a small subset of that experimental logic, but add parameter count as the main scaling axis. This version drops the MNIST result and focuses on harder settings: FashionMNIST, SVHN, STL-10, CIFAR-10, corrupted CIFAR-10 evaluation, and a longer CIFAR-10 patch-transformer schedule. We compare accuracy, pairwise prediction agreement, Jensen-Shannon divergence where available, ensemble gain, and linear interpolation barriers.

1 Background

Let a neural network with parameters θ define a predictive distribution $p_\theta(y \mid x)$ and a class prediction

$$f_\theta(x) = \arg \max_y p_\theta(y \mid x).$$

Deep ensembles train several networks from different random initializations, producing parameters $\theta_1, \dots, \theta_M$. The ensemble prediction is usually the arithmetic mean of predictive distributions:

$$p_{\text{ens}}(y \mid x) = \frac{1}{M} \sum_{m=1}^M p_{\theta_m}(y \mid x).$$

If the ensemble members make accurate but decorrelated errors, averaging can improve both accuracy and uncertainty estimates.

The key distinction in Fort et al. is between *weight-space* similarity and *function-space* similarity. Two models may be far apart in weight space but compute similar functions, or they may be connected by a low-loss path while still representing substantially different classifiers. The paper therefore measures not only loss or parameter distance, but also prediction-space quantities such as disagreement, Jensen-Shannon divergence, and t-SNE embeddings of softmax vectors.

2 Connection to Fort et al.

Fort et al. perform several experiments that motivate this project:

- compare checkpoints within one training trajectory against checkpoints from different random initializations;
- visualize prediction vectors with t-SNE;
- sample local subspaces around one optimum using dropout, diagonal Gaussian approximations, low-rank Gaussian approximations, and random subspaces;
- build diversity-accuracy plots comparing independent initializations to subspace samples;
- compare baseline models, subspace sampling, deep ensembles, and ensemble-plus-subspace methods;
- test behavior under dataset shift on CIFAR-10-C and ImageNet-C.

This report focuses on a narrower question: when architecture width and parameter count increase, do independently trained minima become more similar as functions? This is related to the paper’s diversity-accuracy plane, but uses parameter count as an explicit independent variable.

3 Metrics

For a pair of independently trained models a, b , the experiments compute:

$$\text{agreement}(a, b) = \frac{1}{N} \sum_{n=1}^N \mathbf{1}\{f_{\theta_a}(x_n) = f_{\theta_b}(x_n)\}, \quad (1)$$

$$\text{disagreement}(a, b) = 1 - \text{agreement}(a, b). \quad (2)$$

The Jensen-Shannon divergence is computed from the predictive distributions:

$$\text{JS}(p_a, p_b) = \frac{1}{2} \text{KL}(p_a \| m) + \frac{1}{2} \text{KL}(p_b \| m), \quad m = \frac{1}{2}(p_a + p_b).$$

We also compute a two-member ensemble accuracy for each pair and subtract the mean individual accuracy to estimate an *ensemble gain*. Finally, to probe loss-landscape connectivity, we linearly interpolate in parameter space:

$$\theta(\alpha) = (1 - \alpha)\theta_a + \alpha\theta_b, \quad \alpha \in [0, 1],$$

and report the maximum loss barrier relative to the worse endpoint loss.

4 Experimental Setup

All runs used PyTorch with Apple’s Metal backend (`mps`) exposed to the process. This report uses completed series experiments rather than the original MNIST pilot. The CNN dataset-difficulty experiments use five random seeds per width and full training sets; the newly added SVHN and STL-10 sweeps use three random seeds per width. The corrupted-evaluation runs train on clean CIFAR-10 and evaluate diversity on noisy or blurred test inputs. The patch-transformer run uses a longer schedule than the initial pilot, but remains an optimization stress test rather than a tuned vision transformer. CIFAR-100 was configured as an additional hard dataset, but is not included in the results because the upstream dataset host returned HTTP 503 before training began.

Experiment	Dataset / Eval	Architecture	Widths
FashionMNIST CNN	FashionMNIST	CNN	16, 32, 64, 128
CIFAR-10 CNN	CIFAR-10	CNN	16, 32, 64, 128
SVHN CNN	SVHN	CNN	16, 32, 64, 128
STL-10 CNN	STL-10	CNN	16, 32, 64, 128
CIFAR-10 Noise Eval	CIFAR-10 + noise	CNN	16, 32, 64, 128, 256
CIFAR-10 Blur Eval	CIFAR-10 + blur	CNN	16, 32, 64, 128, 256
CIFAR-10 Patch Transformer	CIFAR-10	Patch transformer	64, 128, 256, 512

Table 1: Harder non-MNIST experiments included in this report.

5 Results

5.1 Dataset Difficulty: FashionMNIST and CIFAR-10

Dataset	Width	Params	Acc.	Agreement	Ens. Gain	Barrier
FashionMNIST	16	38.3k	90.22%	92.99%	0.76 pp	1.638
FashionMNIST	32	151.3k	91.70%	93.43%	0.81 pp	1.728
FashionMNIST	64	601.6k	91.94%	93.16%	0.95 pp	1.754
FashionMNIST	128	2.40M	92.08%	93.52%	0.80 pp	1.696
CIFAR-10	16	38.6k	70.91%	74.66%	2.90 pp	1.361
CIFAR-10	32	151.9k	74.22%	74.85%	3.13 pp	1.256
CIFAR-10	64	602.8k	75.70%	75.97%	3.03 pp	0.722
CIFAR-10	128	2.40M	75.67%	76.05%	2.90 pp	0.431

Table 2: Dataset-difficulty results. CIFAR-10 preserves much more disagreement and ensemble benefit than FashionMNIST at comparable parameter counts.

The drop from FashionMNIST to CIFAR-10 changes the diversity story. FashionMNIST models are accurate and highly agreement-heavy: pairwise disagreement stays near 6–7%, and the two-member ensemble gain is below one percentage point. CIFAR-10 is harder and keeps substantially more functional diversity: disagreement stays near 24–25%, and ensemble gain is around three percentage points. This is more consistent with the deep-ensemble intuition that harder problems leave room for independent initializations to make complementary errors.

5.2 New Dataset Sweeps: SVHN and STL-10

SVHN reaches high accuracy quickly and behaves closer to FashionMNIST than to CIFAR-10: agreement is high, and ensemble gain is moderate. The widest SVHN run does not improve accuracy, suggesting mild overfitting or optimization instability at fixed schedule. STL-10 is a more useful stress test for this research question. Accuracy is much lower, disagreement stays above 22% even at the widest setting, and interpolation barriers shrink sharply with width. This combination suggests that wider STL-10 CNNs become more linearly connected while still retaining nontrivial function-space diversity.

5.3 Corrupted CIFAR-10 Evaluation

The corrupted evaluations are the strongest evidence that clean in-distribution agreement can hide useful diversity. Under Gaussian noise, accuracy collapses and pairwise agreement falls toward

Dataset	Width	Params	Acc.	Agreement	Ens. Gain	Barrier
SVHN	16	38.6k	87.52%	87.91%	1.80 pp	1.665
SVHN	32	151.9k	89.68%	90.11%	1.42 pp	1.653
SVHN	64	602.8k	89.65%	89.27%	1.83 pp	1.624
SVHN	128	2.40M	88.10%	86.47%	2.44 pp	1.585
STL-10	16	38.6k	56.77%	69.74%	1.97 pp	0.856
STL-10	32	151.9k	60.96%	72.81%	2.03 pp	0.979
STL-10	64	602.8k	62.25%	72.56%	1.78 pp	0.591
STL-10	128	2.40M	63.82%	77.90%	1.25 pp	0.056

Table 3: New completed hard-dataset sweeps. SVHN is easier than CIFAR-10 for this CNN, while STL-10 is harder and preserves more pairwise disagreement.

Eval Variant	Width	Params	Acc.	Agreement	Ens. Gain	Barrier
Gaussian noise	16	38.6k	33.27%	47.97%	1.73 pp	0.000
Gaussian noise	32	151.9k	33.56%	52.32%	0.88 pp	0.000
Gaussian noise	64	602.8k	33.05%	54.00%	0.71 pp	0.000
Gaussian noise	128	2.40M	36.33%	46.82%	2.16 pp	0.000
Gaussian noise	256	9.59M	32.88%	43.93%	1.46 pp	0.000
Blur	16	38.6k	57.96%	59.23%	5.22 pp	1.051
Blur	32	151.9k	60.02%	57.62%	5.52 pp	0.037
Blur	64	602.8k	61.01%	61.85%	4.68 pp	0.000
Blur	128	2.40M	59.93%	59.76%	4.98 pp	0.000
Blur	256	9.59M	55.51%	54.29%	5.48 pp	0.000

Table 4: CIFAR-10 corruption evaluation. Dataset shift sharply reduces agreement and increases the value of averaging predictions, especially under blur.

coin-flip levels for several widths. Under blur, accuracy remains better, but disagreement is much higher than on clean CIFAR-10 and ensemble gain rises to roughly five percentage points. The near-zero interpolation barriers in several corrupted rows should be interpreted carefully: the barrier is measured on corrupted evaluation loss, not on the original training objective.

5.4 Longer CIFAR-10 Patch Transformer

The longer patch-transformer schedule improves the smallest model substantially relative to the pilot, but scaling width still does not produce a clean accuracy or similarity improvement. Larger widths become less accurate and less agreement-heavy, with the 512-width model behaving like an undertrained model. This suggests that architecture family and optimization quality dominate any simple parameter-count effect in this regime.

6 Paper-Style Plots

Figure 1 repeats the spirit of Fort et al.’s diversity-accuracy plane for the harder non-MNIST runs. FashionMNIST and SVHN occupy relatively high-accuracy, low-diversity regions; clean CIFAR-10 and STL-10 move toward lower accuracy and higher diversity; corrupted CIFAR-10 is much more

Width	Params	Acc.	Agreement	JS	Barrier
64	208.1k	72.60%	74.12%	0.1001	2.175
128	809.4k	71.32%	70.71%	0.1426	2.488
256	3.19M	66.42%	63.96%	0.1859	1.670
512	12.67M	46.28%	43.52%	0.1472	0.536

Table 5: Longer CIFAR-10 patch-transformer schedule. Wider transformers remain hard to optimize in this setup and become less similar in function space.

diverse but also less accurate.

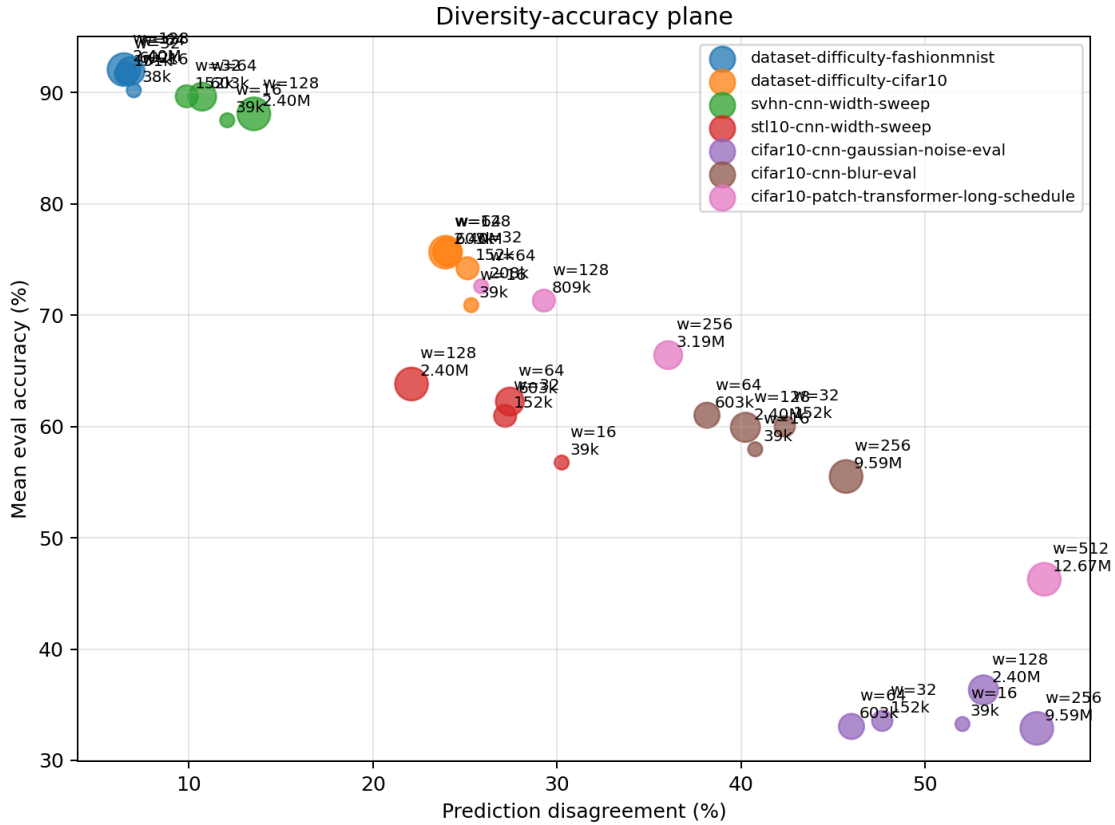


Figure 1: Diversity-accuracy plane for harder non-MNIST runs. Marker size and labels are tied to parameter count.

7 Interpretation

Dropping MNIST makes the conclusion less optimistic for a simple “wider means same function” hypothesis. Dataset difficulty and evaluation shift matter at least as much as parameter count.

- FashionMNIST is easy enough that independent CNNs converge to highly similar predictors, with little ensemble benefit.
- SVHN is also high-accuracy in this setup, though the widest model loses some accuracy and

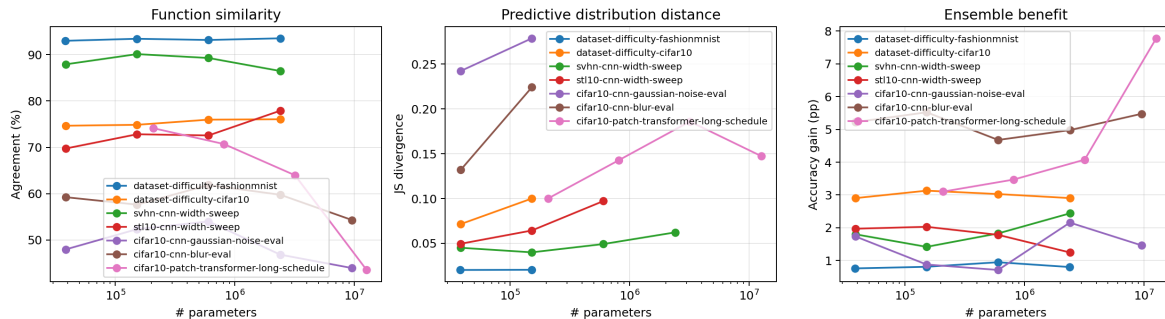


Figure 2: Function similarity, predictive divergence, and ensemble gain as a function of parameter count. Some cached JS-divergence rows contain missing values, so the most reliable comparisons are agreement and ensemble gain.

agreement under the fixed training schedule.

- Clean CIFAR-10 preserves much more diversity and meaningful ensemble gain even as width increases.
- STL-10 is harder than SVHN and preserves substantial disagreement, while its interpolation barriers shrink strongly at the widest setting.
- Corrupted CIFAR-10 exposes still more disagreement, especially under blur, showing that diversity can reappear under dataset shift.
- The patch transformer remains optimization-limited: width increases parameters, but does not automatically improve accuracy or collapse functional diversity.

8 Next Experiments

The next version should make the harder-data claims more robust:

1. Fix the numerical issue that produced missing JS-divergence rows in some cached pairwise metrics by clamping predictive distributions before KL computations, then recompute pairwise CSVs.
2. Retry CIFAR-100 once the upstream dataset host is available, or add a maintained mirror so downloads do not block the experiment.
3. Add true CIFAR-10-C severities rather than only one Gaussian-noise and one blur setting.
4. Add Tiny ImageNet as a genuinely different harder dataset once the data pipeline supports non-torchvision archives.
5. Train all widths to matched endpoint train loss to separate width from optimization quality.
6. Tune the transformer with warmup, cosine decay, augmentation, and longer schedules before treating it as evidence about architecture scaling.

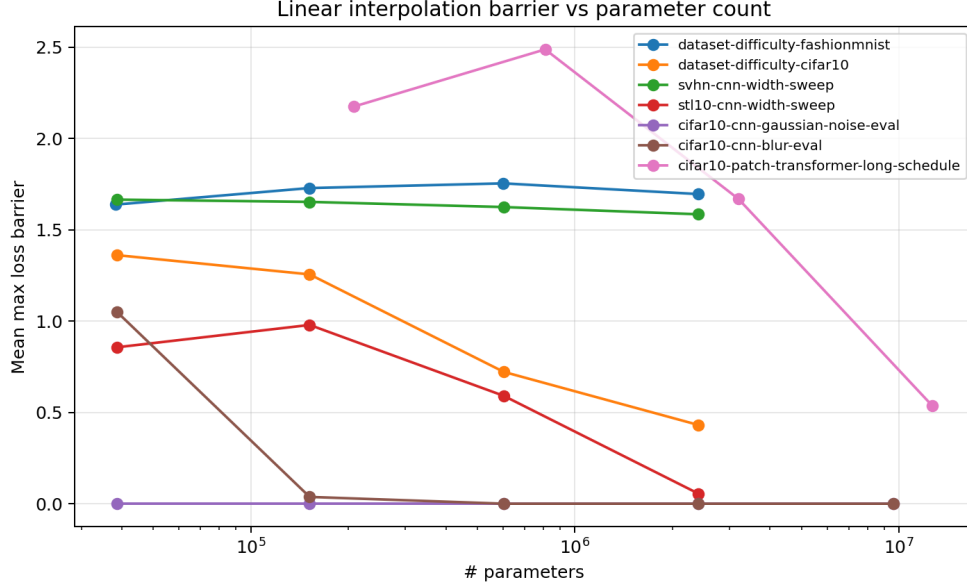


Figure 3: Mean maximum linear interpolation barrier versus parameter count.

9 Conclusion

The harder non-MNIST experiments suggest that functional similarity does not collapse universally with parameter count. FashionMNIST and SVHN already leave less room for ensemble diversity, while CIFAR-10, STL-10, and corrupted CIFAR-10 preserve substantial disagreement and ensemble gain. The strongest current conclusion is that overparameterization interacts with task difficulty, shift, and optimization. Wider models can become more similar or more linearly connected, but harder data can keep independent minima functionally distinct enough for ensembles to matter.